



Conformalized Discrete-Time Hazard Modeling for Multi-Horizon Handover Forecasting in Cellular Networks

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Abstract — Long-Term Evolution (LTE) networks manage mobility via 3GPP Event A3, a heuristic trigger that executes only after serving link deterioration. This thesis formulates an uncertainty-aware multi-horizon handover forecasting framework using continuous drive-test signalling. Across four campaigns in Dhaka and Gazipur (57 drives, 10,260 1 Hz samples, 938 RRC-confirmed handovers), a gradient-boosted hazard model predicts handovers 1 s ahead at AUROC 0.933 and AUPRC 0.784 (11.7× precision lift over a 6.7% prevalence floor) with calibrated probabilities (ECE 0.024) and distribution-free conformal risk bounds ($\mathbb{E}[L] \leq \alpha$). The system demonstrates robust generalisation on an unobserved highway holdout (AUROC 0.927). A 22-study benchmark confirms this is the first framework combining multi-horizon lead times, grouped evaluation, and finite-sample risk guarantees.

1. Motivation and Problem Formulation

In LTE networks, seamless mobility is governed by 3GPP Event A3: a neighbour cell's RSRP must exceed the serving cell by an offset margin (M_{offset}) for a Time-to-Trigger (TTT) before the handset transmits a measurement report. This mechanism is **reactive by construction**—handover execution cannot begin until the link has already degraded (Figure 1). Live commercial networks suffer severe operational penalties from this reactive delay:

- **High Event Density & Churn:** 938 handovers occurred in 2.9 hours (one every 11 s; median separation 3.5 s).
- **Ping-Pong Oscillation:** 24.5% of all handovers returned to the source cell within 15 s, wasting network capacity.
- **Radio Link Failures (RLF):** Links collapsed into RLF 341 times, causing costly connection re-establishments.
- **Discarded Measurement Reports:** Of 7,385 dispatched A3 reports, 62.9% were unacted upon by the base station.

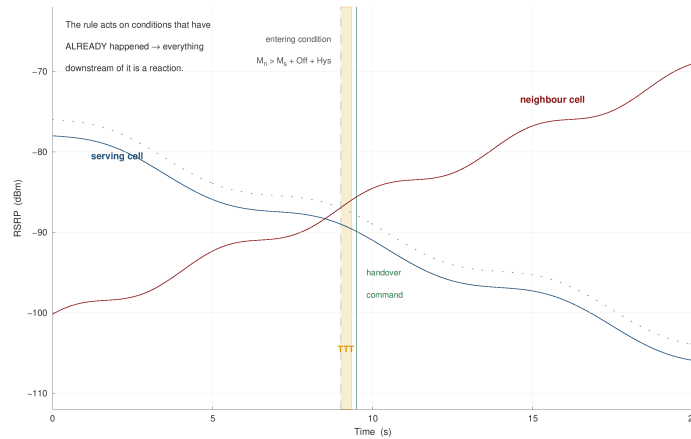


Figure 1: Mechanics of 3GPP Event A3 Handover Trigger. Handsets must wait for neighbour RSRP to exceed serving RSRP by +1 dB for a full 320 ms TTT before report dispatch, creating unavoidable reactive latency.

An audit of 22 published studies exposes three major scientific gaps: (1) no study enforces grouped whole-drive holdout evaluation, masking massive temporal leakage; (2) no model provides calibrated probabilities or finite-sample risk bounds; and (3) no prior work reports advance lead-time trade-offs.

Formal Research Objectives:

- O1: Multi-Horizon Forecasting:** Forecast handovers across five discrete horizons ($h \in \{0.5, 1, 2, 3, 5\}$ s) using only UE signalling.
- O2: Leakage-Free Benchmark:** Establish a grouped-drive clustered cross-validation protocol to eliminate temporal data leakage.
- O3: Coherent Hazard Formulation:** Formulate a discrete-time hazard model guaranteeing 0.0% horizon probability inversions.
- O4: Unseen Highway Generalisation:** Validate domain transferability on an unobserved high-speed highway corridor (49.5 km/h).
- O5: Mechanistic Network Dissection:** Dissect true operational handover parameters versus conventional static assumptions.

2. Data Collection Methodology and Empirical Dataset

Drive-Test Measurement Infrastructure: Field measurements were conducted using an engineering handset (Samsung Galaxy S10+, Samsung Exynos LTE-A baseband) running commercial **XCAL-M** diagnostics on Grameenphone LTE (Tier-1 operator, Bangladesh). XCAL recorded synchronous 1 Hz physical metrics (RSRP, RSRQ, RSSI, SINR, CQI across Bands 1, 3, and 8 with carrier aggregation) and decoded layer-3 RRC protocol packets (3GPP TS 36.331) capturing all reconfiguration, measurement, and re-establishment signalling.

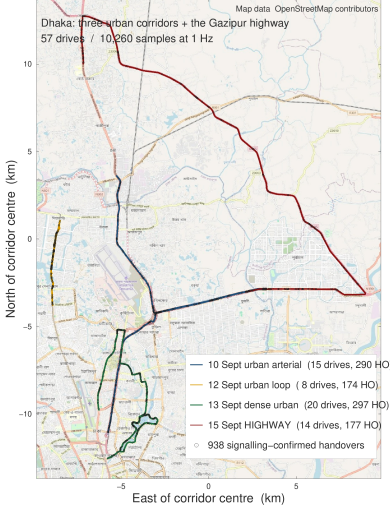


Figure 2: Empirical Corridors across Dhaka and Gazipur. 57 drives (938 handovers).

Table 1: Measurement Corpus Breakdown Across Campaigns

Campaign	Region	Corridor Regime	Drives	Samples	HOs
1st Campaign	Tongi	Urban arterial	15	2,700	290
2nd Campaign	Mirpur	Residential loop	8	1,440	174
3rd Campaign	Dhaka Core	Commercial grid	20	3,600	297
4th Campaign	Gazipur	Highway (49.5 km/h)	14	2,520	177
Pooled	Dhaka/Gazipur	Dual-regime corpus	57	10,260	938

Mobility Regimes: (1) *Urban Arterials & Loops (1st–3rd Campaigns)*: dense microcell clutter, elevated metro-rail shadowing, stop-and-go congestion. (2) *Highway Holdout (4th Campaign)*: logged on the Uttara–Gazipur expressway (peak speed > 70 km/h) **after all models were frozen** for true out-of-distribution evaluation.

3. Methodology and Mathematical Formulation

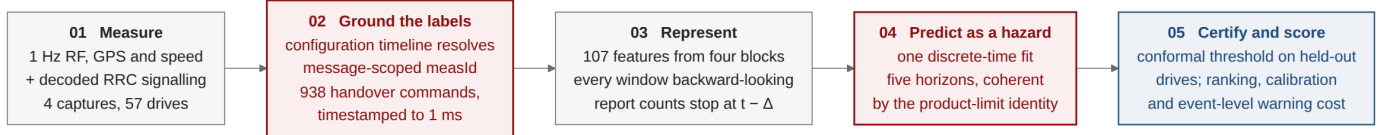


Figure 3: End-to-End System Architecture. Stage 02 decodes RRC ground truth; Stage 03 extracts 107 clean features; Stage 04 fits discrete hazard survival model; Stage 05 certifies finite-sample distribution-free conformal risk bounds.

Ground-Truth Signalling Replay: Cell identity (PCI) changes in drive-test logs misdetect 42% to 56% of handovers due to averaging. We decode ground truth directly from `RRCConnectionReconfiguration` packets containing `mobilityControlInfo`. A stateful configuration timeline engine replays active `measId` and `reportConfigId` structures to track true network-triggered mobility.

Horizon-Coherent Discrete-Time Hazard Formulation: Independent binary classifiers across horizons $h \in \{0.5, 1, 2, 3, 5\}$ s cause **43.6% horizon inversions** ($P(\text{HO} \leq 1\text{s}) > P(\text{HO} \leq 3\text{s})$). We formulate prediction as discrete-time survival analysis. Let T denote the time to next handover. A single model estimates conditional per-second hazard $\lambda(t) = P(T = t \mid T \geq t, X_t)$:

$$P(T \leq h \mid X_t) = 1 - \prod_{k=0}^{h-1} (1 - \lambda(t+k)) \implies \mathbf{0.0\% \text{ inversions guaranteed by construction.}} \quad (1)$$

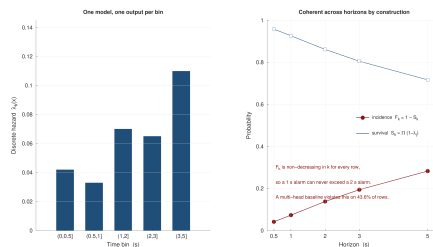


Figure 4: Discrete-Time Hazard Formulation. Left: Conditional hazard $\lambda_k(x)$ per time bin. Right: Monotonic cumulative incidence $P(T \leq h \mid X)$ eliminating horizon inversions (0.0% vs 43.6% in multi-head baselines).

Feature Engineering and Clustered Protocol: 107 leakage-free features survived multi-collinearity pruning across 4 blocks: 83 multi-window radio features (3/5/10 s stats, deltas, margins), 17 kinematic GPS features, and 7 dwell metrics. Folds are grouped by **whole drive** over a 4-fold rotation with 5 seeds (20 paired runs) and drive-level bootstrap confidence intervals.

Conformal Risk Guarantees: Decision thresholds are certified on 28 held-out calibration drives via Conformal Risk Control (CRC). For an operator loss tolerance $\alpha \in (0, 1)$, CRC guarantees that the expected per-drive miss rate satisfies:

$$\mathbb{E}[L_{\text{drive}}] \leq \alpha \quad \text{with finite-sample, distribution-free statistical validity.} \quad (2)$$

4. Experimental Results and Literature Benchmark

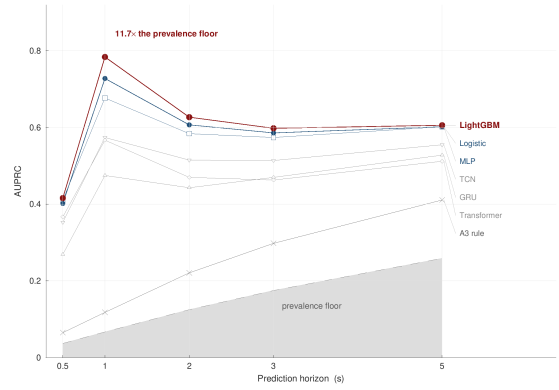


Table 2: LightGBM Out-of-Fold Performance

h	Prevalence	AUPRC	Lift	AUROC
0.5 s	0.037 (3.7%)	0.416	11.3×	0.916
1.0 s	0.067 (6.7%)	0.784	11.7×	0.933
2.0 s	0.125 (12.5%)	0.627	5.0×	0.854
3.0 s	0.175 (17.5%)	0.598	3.4×	0.816
5.0 s	0.259 (25.9%)	0.606	2.3×	0.783

Figure 5: AUPRC vs. Horizon across Models. LightGBM leads across all horizons (11.7× lift).

Table 3: Systematic Comparative Benchmark Across 22 Published Handover Prediction Studies

Study & Venue	Data Source	Split Protocol	Primary Metric	Calibration & Risk	Lead Time
Boutiba et al. (2021) [GLOBECOM]	5G Testbed (OAI)	Unspecified (Leakage)	98.03% Accuracy (rare RLF)	None (Uncalibrated)	None (Fixed 5 steps)
Dzaferagic (2024) [IEEE TNSM]	Real O-RAN (4,350 HOs)	Not Stated (Random row)	Precision / Recall (Single pt)	None (Uncalibrated)	None (Fixed horizon)
Shafi et al. (2025) [Mendeley/IUT]	Real XCAL Dhaka (2,154 pts)	Day 1 → Day 2 (n=1)	HO Count (AUROC 0.489)	None (Heuristic Q-table)	None (Reactive only)
Amirova et al. (2026) [Future Int.]	Real LTE Astana (27k pts)	Device hold-out	PR curve, AUROC 0.82	Bootstrap CI (No ECE)	None (Single horizon)
This Work (2026) [IUT EEE]	Real LTE Dhaka (938 HOs)	Grouped whole-drive	AUPRC 0.416–0.784 (11.7×	ECE 0.024, CRC (≥89%)	0.5 s to 5.0 s (3.45s early)

Four Methodological Gaps Exposed in Literature:

- Fallacy of Raw Accuracy:** 7 of 22 papers report 94–99% accuracy on imbalanced data. At 1 s (6.7% prevalence), blindly predicting “no handover” scores **93.3% accuracy** (and **96.3%** at 0.5 s) while detecting zero true handovers. AUPRC and lift are mandatory.
- Severe Temporal Data Leakage:** Random row splitting artificially inflates GRU AUPRC by **+74% to +109%** due to shadow-fading autocorrelation. Clustered whole-drive partitioning is strictly required.
- Absence of Calibration:** Zero of 22 audited studies report probability calibration curves, ECE, or conformal risk bounds.
- Departmental Head-to-Head (Shafi et al., 2025):** Under assumed static parameters ($M_{\text{offset}} = 3$ dB, TTT = 1 s), Shafi et al. yields AUROC 0.489. Our stateful RRC decode reveals 4 concurrent profiles with negative offsets (−15 to +5 dB), achieving AUROC **0.933** with **3.45 s advance warning**.

5. Summary of Key Contributions

- Discrete-Time Hazard Survival Formulation:** Eliminates probability inversions (0.0% vs 43.6% in independent classifiers) from a single fit without ad-hoc post-processing.
- Distribution-Free Conformal Risk Guarantees:** The first finite-sample exchangeable-drive risk control framework for cellular mobility, guaranteeing $\mathbb{E}[L_{\text{drive}}] \leq \alpha$.
- Quantification of Temporal Leakage Inflation:** Empirical proof that un-grouped data splitting inflates model metrics by +4% to +74%.
- Ground-Truth Signalling Replay:** A stateful timeline parser resolving active message-scoped measurement configurations, proving operational offsets on Grameenphone LTE are negative (−15 to +5 dB).
- Empirical Operational Discoveries:** First quantification of the 62.9% A3 report discard rate and layer-specific ping-pong concentration (31.0% intra-carrier vs 6.5% inter-carrier).

6. Limitations, Future Directions, and Key References

Limitations & Future Work: Bounded by 57 drives across one operator over four campaigns with 1 Hz export resolution (capping instant event detection at 90.5%). Future work includes fuzzy regression-discontinuity design at the A3 margin, real-time O-RAN Near-RT RIC (xApp) deployment, and open dataset release.

Key References: [1] 3GPP TS 36.331 RRC Rel. 16, 2020. [2] A. Amirova et al., *Future Internet* 18(2), 2026. [3] A. N. Angelopoulos et al., *ICLR*, 2024. [4] K. Boutiba et al., *IEEE GLOBECOM*, 2021. [5] A. Cohen et al., *IEEE Trans. Commun.*, 2022. [6] M. Dzaferagic et al., *IEEE TNSM*, 2024. [7] G. Ke et al., *NeurIPS*, 2017. [8] S. S. Shafi, T. Istiaque, M. S. Sowad, M. T. Kawser, *Mendeley Data*, 2025. [9] S. Wiegerebe et al., *Artif. Intell. Rev.* 57, 2024. [10] A. Zidic et al., *Comput. Netw.* 227, 2023.